```
Your goal is to bring the Agent closer to the Goal as
    efficiently as possible.
Only respond with the JSON object in the exact format
    specified, using one of the four allowed action strings.
```

Examples of SIRs are provided in the right panels of Fig. 1.

The full code to build and execute the GWSOT, as well as for all statistical analysis, is open source and available on GitHub.

B Summary of Experiments

In this appendix, we provide a detailed summary of all experiments conducted for the Grid-World Spatial Orientation Task (GWSOT). For each experiment, we specify the models used, the spatial information representations (SIRs) applied, and the number of trials run under each condition. A random policy agent was also evaluated as a baseline. When analyzing experimental results, a statistical threshold of 0.05 was used throughout this study.

B.1 Standard GWSOT Across Model Sizes

We evaluated six LLaMA-3 models (3.2-1B, 3.2-3B, 3.1-8B, 3.2-11B, 3.1-70B, and 3.2-90B) on the standard 5×5 grid. For each model, every single SIR was tested individually.

Table 1. Summary of Experiment 1

Representation	Number of Trials per Model
JSON	100
Chess Notation	100
Symbol Grid	100
Word Grid	100
Row Description	100
Column Description	100
Random Policy	100 (overall)

B.2 Probing

For probing analyses, we focused on the LLaMA-3.1-8B model using all SIR conditions. The final dataset comprised individual world-state/action pairs (between 3 and 10 per trial) along with their corresponding hidden-state activations.

Table 2. Summary of Experiment 2

Representation	Number of Trials per Model
JSON	50
Chess Notation	50
Symbol Grid	50
Word Grid	50
Row Description	50
Column Description	50

B.3 Ablation Studies

We conducted ablation experiments on the LLaMA-3.1-8B model using only the JSON SIR. We silenced either units correlated with action correctness across all SIRs or units correlated with agent position across all SIRs. We also evaluated the effect of silencing all units correlated with action correctness only in JSON SIR trials, but this disrupted basic output coherence, so we only conducted a small number of trials for this condition.

Table 3. Summary of Ablation Experiments

Ablation Condition	Number of Trials per Model
No Ablation (Control)	100
Ablation of Core Action-Correctness Units	100
Ablation of Core Agent-Position Units	50
Ablation of All Units Correlated with JSON SIR Action-Correctness	10

C Linear Probing and Identification of Feature-Predicting Units

In this appendix, we describe in detail the methods used to probe internal representations of spatial information in the LLaMA-3.1-8B model. Our analyses include (1) training linear regression models to decode the full grid configuration, (2) analyzing individual neurons for significant correlations with specific spatial features, (3) determining common significant units across different SIR types, and (4) silencing groups of units in ablation experiments while evaluating the model on a GWSOT.

C.1 Linear Regression for Grid Configuration

For each SIR, we first extracted activations from every layer of the model. Activations were averaged over all input tokens and then used to predict the complete configuration of the grid. Specifically, the target was a 50-dimensional binary vector—25 dimensions indicating the presence (1) or absence (0) of the agent and 25 for the goal. For each SIR and for each layer separately, a standard linear regression model was trained on 90% of the data (training set) and evaluated on the remaining 10% (test set). A total of (layers $\times$ SIRs) $= 32 \times 6 = 192$ models were trained. The performance of these linear models was quantified by the coefficient of determination (R^2), both for the training and the test set. Training set R^2 was always 1.0 for every model we trained.

To assess the invariance of the internal representations across different SIRs, we performed cross prediction experiments. Linear models trained on activations from one SIR were evaluated on the test sets of each of the other SIRs separately (i.e., each model was evaluated on six test sets, always receiving activations from the layer they were trained on). R^2 scores were computed for each evaluation and averaged across layers when applicable.

For each layer, we created a null baseline by shuffling the target values (i.e., permuting Y_{train}) to break the true relationship between activations and grid configuration. The null models consistently produced highly negative R^2 values (i.e., extremely poor predictive performance), confirming that the true models captured meaningful spatial information.

C.2 Analysis of Individual Units

To identify which units are predictive of specific spatial features, we fitted a simple linear model for each model parameter using each parameter's activations when producing the first decision token to predict the spatial feature we cared about. In these analyses, the length of the predictor vector X and the target vector Y was the total number of world-state/action pairs in the dataset of a specific SIR type. X was built from parameter activations in each step and Y contained the spatial feature of interest (e.g., a binary indicator of whether the agent is in a particular grid location, or near the grid's border, etc.). P-values for each unit were adjusted by using a Bonferroni correction to account for multiple comparisons (usually by $n_{\text{layers}} \times n_{\text{params\_per\_layer}}$, except when doing comparisons for every grid cell, where the number of comparisons was $5 \times 5 \times n_{\text{layers}} \times n_{\text{params\_per\_layer}}$). A significance threshold of 0.05 was used to determine whether a specific parameter was significantly correlated with the spatial feature of interest after correction.

C.3 Identification of Common Significant Units

To isolate units that reliably encode spatial features regardless of the representation, we determined common units across all SIRs. For each SIR, a binary mask was created indicating whether a parameter's corrected p-value (after Bonferroni adjustment) was below the significance threshold. Parameters that were significant for all SIRs were identified by taking the logical AND across the masks for all SIRs. We compare the number of parameters included in this mask with a shuffle control, where each layer's parameter indices were randomly permuted between SIRs.